

Answer Key — Core Mathematics Grade 10

Sub-Strand 2.4: Trigonometry 1 • Lesson 8: Synthesis, Real-Life Applications, and DQB Review — Solving the Nakuru Mast at Last

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	A	B	D	A	B	D	A

USE AFTER: PREDICT PHASE — Predict Phase: writing the Nakuru mast calculation $\tan 50^\circ = \text{height} \div 40 \text{ m}$ before class sharing

Q1. A student stands 40 m from the base of a mast and measures the angle of elevation to the top as 50° . The 40 m side is adjacent to the 50° angle, and the height of the mast is the opposite side. Which ratio should be used to find the height?

- A. $\sin 50^\circ = \text{height} \div 40$
- B. $\cos 50^\circ = \text{height} \div 40$
- C. $\tan 50^\circ = \text{height} \div 40$**
- D. Pythagoras' theorem only, since 40 m is a known side

Correct answer: C. The known side is adjacent and the unknown side is opposite, and tangent links opposite and adjacent. Sine and cosine involve the hypotenuse, which is not known, and Pythagoras needs two sides.

USE AFTER: OBSERVE PHASE — Observe Phase: STEP 4 — Solve, $h = 40 \times \tan 50^\circ$ for the Nakuru mast

Q2. A mast stands on level ground. From a point 40 m from its base, the angle of elevation to the top is 50° . Given that $\tan 50^\circ \approx 1.1918$, what is the height of the mast to the nearest 0.1 m?

- A. 47.7 m**
- B. 33.6 m
- C. 30.6 m
- D. 25.7 m

Correct answer: A. $\tan 50^\circ = h \div 40$, so $h = 40 \times \tan 50^\circ \approx 40 \times 1.1918 \approx 47.7 \text{ m}$. Dividing by $\tan 50^\circ$ gives 33.6 m, which is a rearrangement error.

USE AFTER: OBSERVE PHASE — Observe Phase: STEP 2 — Identify known and unknown sides relative to the 50° angle

Q3. In the right-angled triangle for the mast problem, the observer's foot is at the 50° angle. The mast is vertical, the ground distance to the mast is 40 m, and the observer looks along a line of sight to the top of the mast. Which statement correctly names the sides relative to the 50° angle?

- A. The 40 m ground distance is the hypotenuse, and the line of sight is adjacent
- B. The 40 m ground distance is adjacent, the mast height is opposite, and the line of sight is the hypotenuse**
- C. The 40 m ground distance is opposite, and the mast height is adjacent
- D. The line of sight is opposite, and the mast height is the hypotenuse

Correct answer: B. Relative to the 50° angle, the side touching the angle along the ground is adjacent and the side across from it (the mast) is opposite. The line of sight is the longest side, the hypotenuse, and it is not needed in this problem.

USE AFTER: EXPLAIN PHASE — Explain Phase: Lake Victoria lighthouse task, drawing the angles of elevation and depression correctly

Q4. A fisherman in a boat on Lake Victoria looks up at the top of a lighthouse and measures the angle of elevation as 30° . A coastguard at the top of the lighthouse looks down at the same boat. Assuming the water level is horizontal, what is the angle of depression from the coastguard to the boat?

- A. 60°
- B. 150°
- C. 90°
- D. 30°**

Correct answer: D. The horizontal line at the top of the lighthouse and the water level are parallel, so the angle of depression equals the angle of elevation (alternate angles). 60° is the complementary angle, which is a common mix-up.

USE AFTER: EXPLAIN PHASE — Explain Phase: Lake Victoria problem, mark scheme step (a) $h_1 = 120 \times \tan 38^\circ$

Q5. A boat on Lake Victoria is 120 m from the base of a vertical cliff. The angle of elevation of the cliff top from the boat is 38° . Given that $\tan 38^\circ \approx 0.7813$, what is the height of the cliff above the water to the nearest 0.1 m?

- A. 93.8 m**
- B. 153.6 m
- C. 73.9 m
- D. 94.6 m

Correct answer: A. The 120 m side is adjacent and the height is opposite, so $\tan 38^\circ = h \div 120$. Then $h = 120 \times 0.7813 \approx 93.8$ m. Dividing 120 by 0.7813 gives 153.6 m, which is the wrong rearrangement.

USE AFTER: EXPLAIN PHASE — Explain Phase: the 'productive contradiction' discussion when the arithmetic $48.5 - 93.8$ does not make sense

Q6. While solving a two-triangle height problem, a learner subtracts two heights to find a cliff height and gets a negative answer. What is the best next step?

- A. Assume the calculator is faulty and guess the height
- B. Redraw the diagram and check which angle belongs to which height, since a length cannot be negative**
- C. Switch to cosine for every part of the problem
- D. Ignore the minus sign and give the answer as a positive number

Correct answer: B. A negative length is impossible, so it signals a set-up error. The lesson's message is that careful diagram work, checking which angle links to which side, fixes such errors.

USE AFTER: DQB PHASE — DQB Review: answering the unit driving question card in one sentence

Q7. The unit driving question asks: 'How can we use the angles and one side of a right-angled triangle to find any unknown height or distance?' Which answer is best?

- A. Use only the tangent ratio in every problem, whatever sides are known
- B. Measure the height directly by climbing with a ladder
- C. Use Pythagoras' theorem, because it links angles to sides
- D. Choose sine, cosine or tangent depending on which sides are known and unknown, then link the known angle and known side to the unknown side**

Correct answer: D. Trigonometric ratios link an angle to a ratio of sides. The ratio chosen depends on the sides involved (SOH-CAH-TOA). Pythagoras relates sides only, and tangent is not the only ratio.

USE AFTER: MODEL BUILDING PHASE — Model Building Phase: annotating the Final Model with the complementary-angle relationship $\sin \theta = \cos(90^\circ - \theta)$

Q8. In their final model, a learner writes the complementary-angle relationship $\sin 35^\circ = \cos \theta$. What is the value of θ ?

- A. 55°
- B. 35°
- C. 145°
- D. 65°

Correct answer: A. The relationship is $\sin \theta = \cos(90^\circ - \theta)$, so $\sin 35^\circ = \cos(90^\circ - 35^\circ) = \cos 55^\circ$.